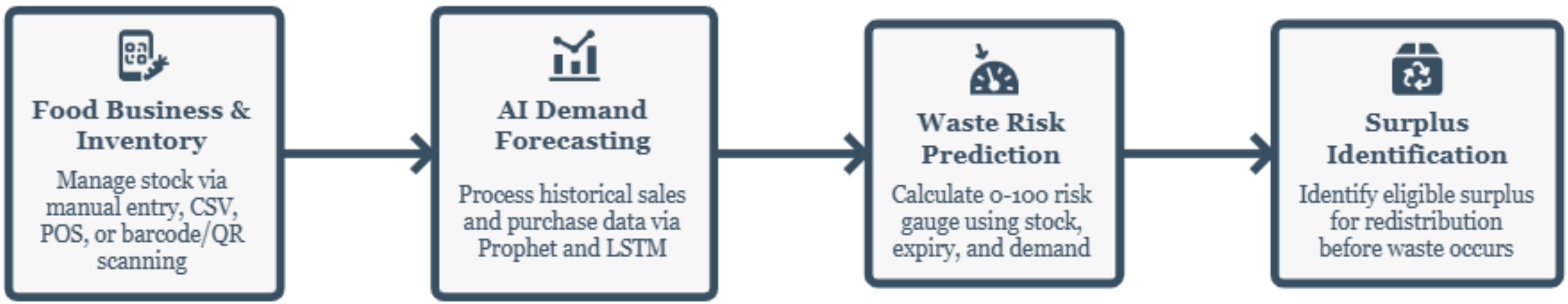
# AI-Based Food Redistribution System



From Waste Prediction to Sustainable Redistribution

Made with Napkin

# AI-Based Food Redistribution: PREDICT Phase

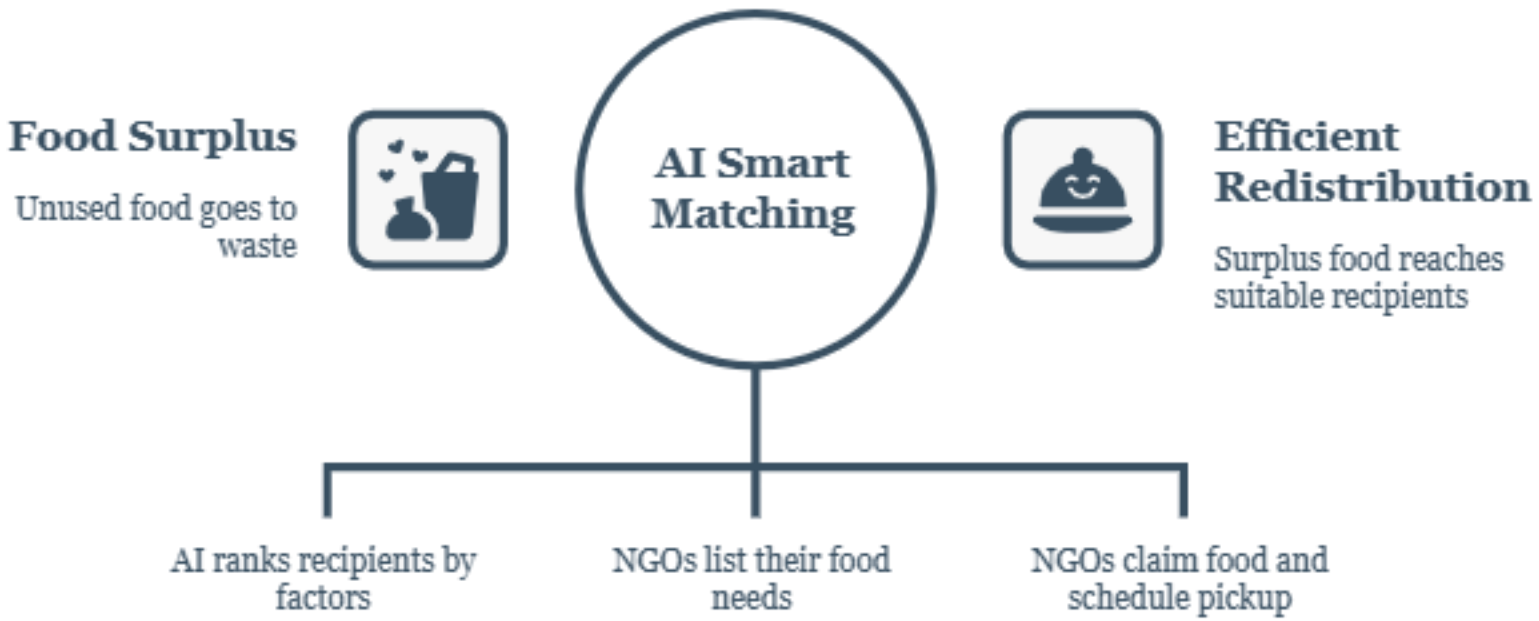



Made with Napkin

# REDISTRIBUTE

Our AI engine ranks suitable recipients based on four key implemented factors:

Food Category, Quantity, Proximity, and Urgency/Expiry.




Made with Napkin



# IMPACT

Measuring social and environmental impact across three key metrics.

01

### Pickup & Digital Verification

Secure handover via digital verification codes at the point of collection.

02

### Redistribution Completed

Successful delivery to community kitchens and NGOs.

03

### Sustainability Analytics

Professional dashboard measuring real-world ROI.

04

### Key Metrics

Food Diverted, Meals Supported, and $CO_2$ Avoided.

Made with Napkin

# AI-Powered Food Waste Reduction Platform - Final Architecture

## Frontend (React + Vite + TypeScript)

### User Flow (Frontend Screens)




**Landing Page** ( / ) LandingPage.tsx

**Login Page** ( /login ) LoginPage.tsx

**Register Page** ( /register ) RegisterPage.tsx

**Dashboard** (Business / NGO / Admin) Protected Routes

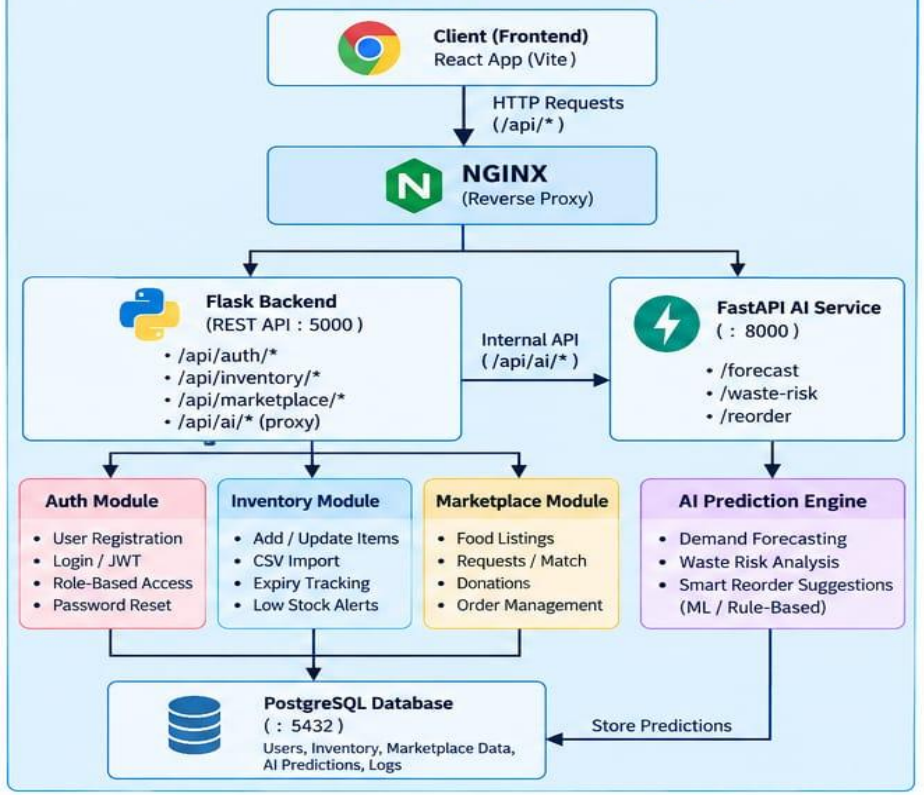
### Frontend Project Structure

```
frontend/
└── src/
    ├── pages/
    │   ├── LandingPage.tsx ← NEW
    │   ├── auth/
    │   │   ├── LoginPage.tsx
    │   │   ├── RegisterPage.tsx
    │   │   ├── ForgotPasswordPage.tsx
    │   │   ├── ResetPasswordPage.tsx
    │   │   └── UnauthorizedPage.tsx
    │   └── dashboard/
    │       ├── BusinessDashboardShell.tsx
    │       ├── NgoDashboardShell.tsx
    │       └── AdminDashboardShell.tsx
    ├── routes/
    │   ├── index.tsx
    │   └── ProtectedRoute.tsx
    ├── context/
    │   └── AuthContext.tsx
    ├── services/
    │   ├── api.ts
    │   ├── auth.ts
    │   ├── inventory.ts
    │   ├── marketplace.ts
    │   └── ai.ts
    ├── App.tsx
    ├── main.tsx
    ├── vite.config.ts
    ├── package.json
    ├── nginx.conf
    └── Dockerfile
```

### Frontend Route Map

| Route | Component |
|---|---|
| / | LandingPage |
| /login | LoginPage |
| /register | RegisterPage |
| /forgot-password | ForgotPasswordPage |
| /reset-password | ResetPasswordPage |
| /unauthorized | UnauthorizedPage |
| /business/dashboard | BusinessDashboardShell (Protected) |
| /ngo/dashboard | NgoDashboardShell (Protected) |
| /admin/dashboard | AdminDashboardShell (Protected) |
| * | Redirect to / |

## Backend (Flask + FastAPI + PostgreSQL)




## System Flow (Complete)

1. Open http://localhost
2. Landing Page ( Sign In / Register )
3. Login / Register ( Auth API )
4. Role-Based Dashboard
5. Use Features (Inventory / AI / Marketplace)

## Testing File

**inventory_test_50_rows.csv**
(50 sample rows for testing)

- Fresh items
- Low-stock products
- Expiring-soon products
- Expired products
- Different categories
- Different suppliers
- Different prices
- Different quantities
- Different storage types
- Barcodes & Expiry dates

## Technology Stack

**Frontend**
React
TypeScript
Vite
Tailwind CSS
Axios
Recharts
Leaflet

**Backend**
Flask
JWT
SQLAlchemy
Marshmallow
Gunicorn

**AI Service**
FastAPI
NumPy
Pandas
Scikit-learn
Uvicorn

**Database**
PostgreSQL

**Deployment**
- Docker
- Docker Compose
- Nginx

**Final Architecture - Ready for Development & Testing**

# TECHNOLOGY STACK

Building a smarter, more sustainable food redistribution platform.

**FoodBridge**
Smart Food Redistribution

Reduce Waste
Feed More People

## Frontend

Modern and responsive web interface for businesses, NGOs and admins.

- React.js
- TypeScript
- Tailwind CSS
- Recharts

## Backend

Robust APIs and secure application services.

- Python
- Flask
- SQLAlchemy
- JWT

## AI / Machine Learning

Intelligent prediction and recommendation models.

- Pandas
- NumPy
- Scikit-learn
- Prophet

## AI Service

Dedicated AI microservice for advanced analytics and predictions.

- FastAPI

## Database

Reliable and scalable data storage for inventory, users and transactions.

- PostgreSQL

## Maps & Location

Find nearby NGOs and optimize pickup routes.

- Leaflet
- React-Leaflet

## Barcode / QR Scanning

Quick and accurate inventory updates using mobile devices.

- html5-qrcode

## Deployment & Testing

Reliable deployment and quality assurance.

- Docker
- Pytest
- Vitest
- Playwright

## Version Control

Source code management and team collaboration.

- Git
- GitHub

Technology for a Better Tomorrow | Smarter Decisions | Less Food Waste | Stronger Communities

**Good Food Brighter Futures**

AI-Powered Food Redistribution

# Reduce Food Waste.
# Feed More People.

An intelligent platform connecting businesses with NGOs to reduce surplus food waste using smart inventory management, AI-powered demand analysis and efficient redistribution.

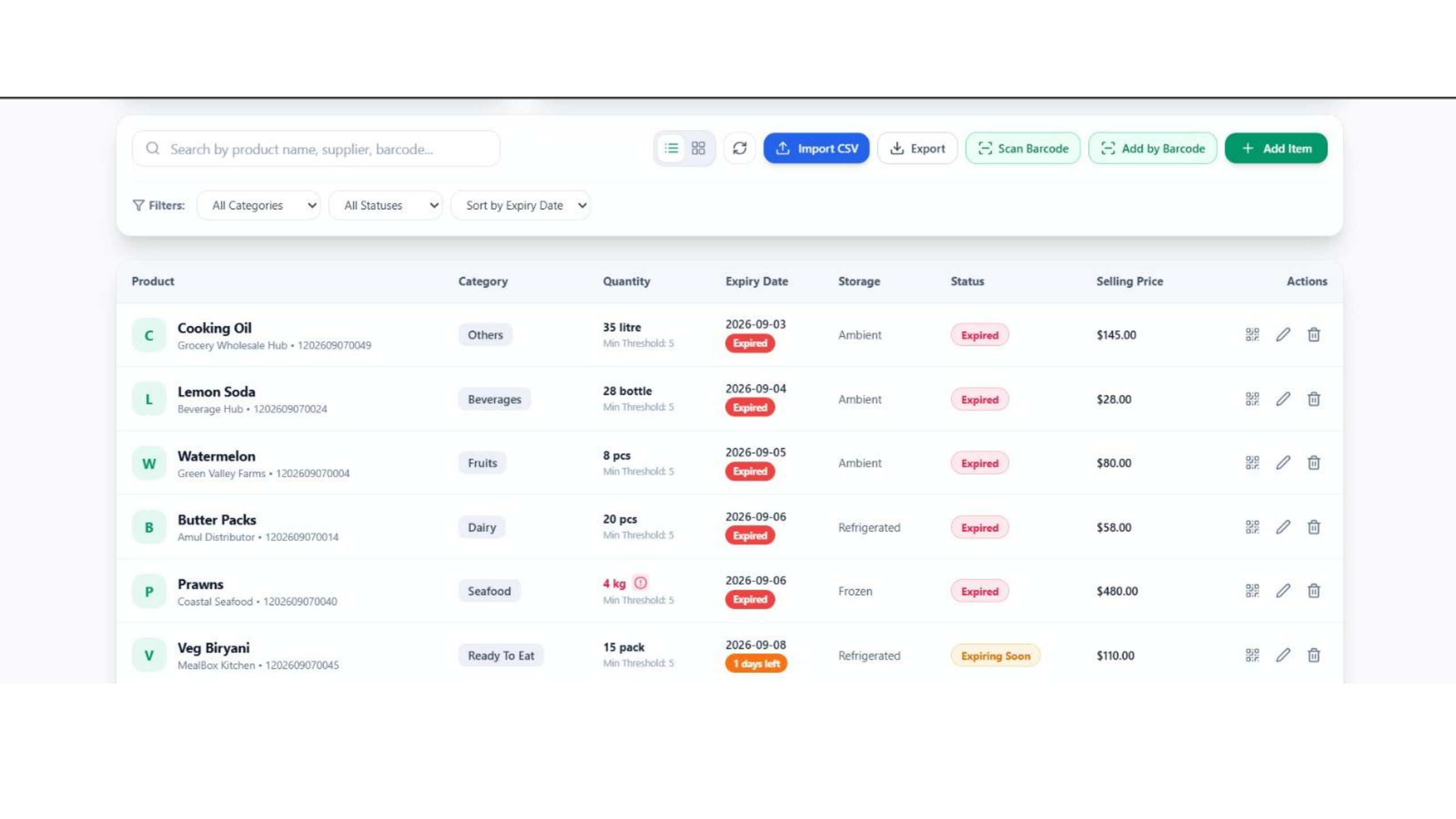
Sign In → Create Account

**AI**
Smart Forecasting

**24/7**
Inventory Tracking

**Live**
Redistribution




Search by product name, supplier, barcode...
Import CSV
Export
Scan Barcode
Add by Barcode
Add Item
Filters:
All Categories
All Statuses
Sort by Expiry Date
Product
Category
Quantity
Expiry Date
Storage
Status
Selling Price
Actions
C
Cooking Oil
Grocery Wholesale Hub • 1202609070049
Others
35 litre
Min Threshold: 5
2026-09-03
Expired
Ambient
Expired
$145.00
L
Lemon Soda
Beverage Hub • 1202609070024
Beverages
28 bottle
Min Threshold: 5
2026-09-04
Expired
Ambient
Expired
$28.00
W
Watermelon
Green Valley Farms • 1202609070004
Fruits
8 pcs
Min Threshold: 5
2026-09-05
Expired
Ambient
Expired
$80.00
B
Butter Packs
Amul Distributor • 1202609070014
Dairy
20 pcs
Min Threshold: 5
2026-09-06
Expired
Refrigerated
Expired
$58.00
P
Prawns
Coastal Seafood • 1202609070040
Seafood
4 kg
Min Threshold: 5
2026-09-06
Expired
Frozen
Expired
$480.00
V
Veg Biryani
MealBox Kitchen • 1202609070045
Ready To Eat
15 pack
Min Threshold: 5
2026-09-08
1 days left
Refrigerated
Expiring Soon
$110.00

## Add New Inventory Item

Fill in the item details, expiry date, and customizable stock threshold.

**Product Name ***

e.g. Organic Milk 1L

**Category ***

Fruits

**Storage Type**

Ambient

**Quantity ***

1

**Measurement Unit ***

units

**Expiry Date ***

dd-mm-yyyy

**Purchase Date**

07-09-2026

**Configurable Low Stock Threshold ***

Default: 5

Triggers a low stock alert when quantity drops to or below this customizable value.

5

**Cost Price ($)**

0

**Selling Price ($)**

0

**ABCD** FOOD BUSINESS

Logged in as **Dipayan Saha** (dipayan0921@gmail.com)

Sign Out

# Inventory & Expiry Control Center

Manage product stocks, monitor expiry risk levels, set custom thresholds, and import/export records.

Phase 3 Active Module

Inventory Stock Items | AI Waste Prediction & Reorders | Redistribution Marketplace | Sustainability Analytics

| Total Inventory Items | Inventory Health | Expiring Soon | Low Stock & Expired |
|---|---|---|---|
| 50 | 47% | 10 | 13 |
| Total Value: $107,104.00 | Status: Critical | Expiring within 3 days | 8 Low Stock • 5 Expired |

## Expiry Timeline

Risk forecasting based on shelf dates

| Today | Tomorrow | Next 7 Days | Next 30 Days |
|---|---|---|---|
| 0 items | 3 items | 19 items | 41 items |

View All Expiring Items

## Urgent Expiry Alerts (10)

Items requiring immediate redistribution or action

- **Mangoes** — 18 kg • Fruits • Expiry: 2026-09-10 — 3 Days Left
- **Carrots** — 15 kg • Vegetables • Expiry: 2026-09-09 — 2 Days Left
- **Paneer** — 4 kg • Dairy • Expiry: 2026-09-08 — 1 Days Left

## Current Risk Distribution

| LOW RISK | MED RISK | HIGH RISK | CRITICAL |
|---|---|---|---|
| 22% | 48% | 6% | 24% |
| 11 | 24 | 3 | 12 |

Total active items analyzed: **50**

## Waste Prevention Score

**Limited historical data**

Need run history for trend metrics

*A composite operational indicator based on actual risk stabilization, redistribution actions, and avoided waste opportunities. It is not model accuracy.*

| | |
|---|---|
| Waste Prevented: | 197 units |
| Financial Savings: | ₹15,170.62 |
| Redistribution Rate: | 19.5% |

## Model Evaluation Metrics

**Insufficient historical data.**

Model evaluation metrics require at least 10 observations.

## AI System Health

Retry

| | | | |
|---|---|---|---|
| Forecast Engine | READY | Risk Scorer | READY |
| Database | READY | FastAPI API | READY |

| | |
|---|---|
| Last Pred Run: | 9/7/2026 |
| Categories Tracked: | 11 categories |

## AI Scheduler Status

ACTIVE

| DAILY CRON | WEEKLY CRON |
|---|---|
| 02:00 AM | Monday 03:00 |

NEXT SCHEDULED PREDICTION RUN
9/8/2026, 7:30:00 AM

RECENT RUNS

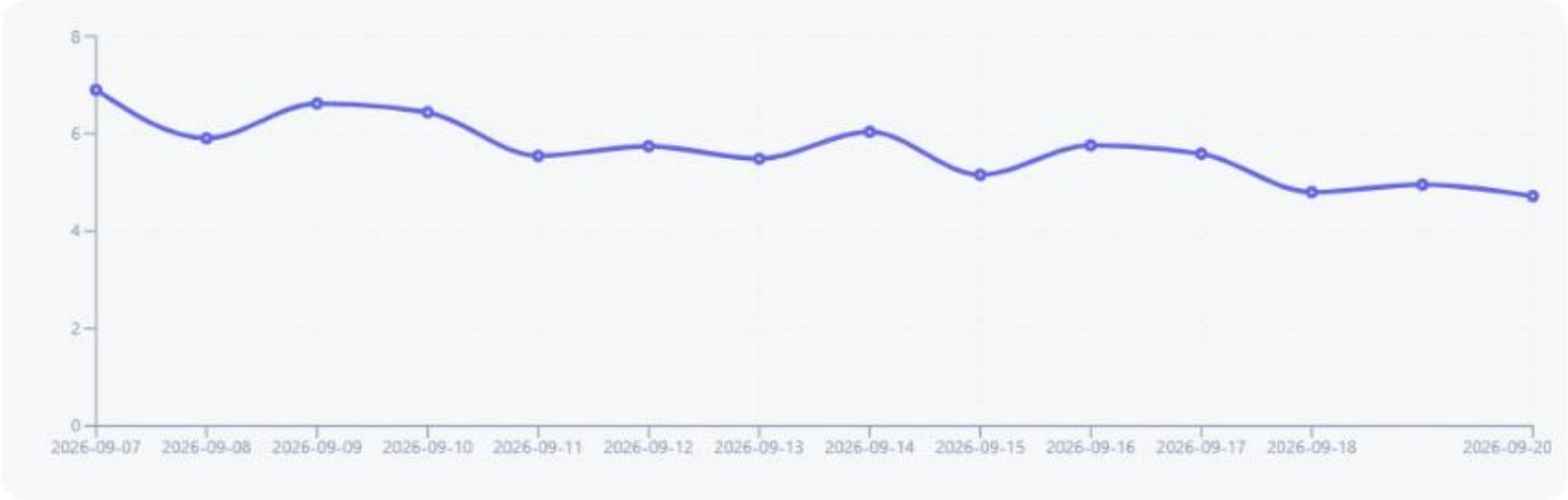
BATCH_PREDICTION_D... COMPLETED
Triggered: Sep 7, 03:45 PM Duration: 0.7s

Run Now | Refresh

## Category Demand Forecast

Trend: DOWNWARD

Dairy | Daily | Weekly




## AI Priority Queue

**Cooking Oil** P1 — 100.0 pts
Risk: 100 | Expiry: EXPIRED 4 days ago | Waste: ₹4200 | EMERGENCY CONSUMPTION

**Prawns** P2 — 100.0 pts
Risk: 100 | Expiry: EXPIRED 1 days ago | Waste: ₹1520 | EMERGENCY CONSUMPTION

**Lemon Soda** P3 — 100.0 pts
Risk: 100 | Expiry: EXPIRED 3 days ago | Waste: ₹504 | EMERGENCY CONSUMPTION

**Butter Packs** P4 — 100.0 pts
Risk: | Expiry: EXPIRED 1 days | Waste: | EMERGENCY CONSUMPTION

# Leanrnng
## What We Have Learnt

FoodBridge
Smart Food Redistribution

Key insights and takeaways from building FoodBridge.

### Real-World Impact
Technology can solve meaningful social and environmental problems.

### Data Matters
Clean, quality data significantly improves AI/ML prediction accuracy and decision-making.

### User-Centric Design
Understanding the needs of NGOs, business owners and volunteers is crucial for adoption.



### Collaboration is Key
Cross-functional teamwork helped us overcome technical and domain challenges.

### Continuous Learning
We gained hands-on experienes with modern tech stacks, AI models, and real-world deployment practices.

### Bigger Perspective
We learned the importance of sustainability and how small actions can create a larger positive impact.

"We didn't just build a system — we learned how technology can create a kinder, more sustainable world."

Learning Today
for a Better Tomorrow

# Feature Enhancements
## Ideas for a Smarter, Greater Impact

FoodBridge
Smart Food Redistribution

Future improvements to make FoodBridge more powerful and scalable.

### More Advanced Predictions
Use deep learning models and real-time data (weather, events, footfall) for even more accurate surplus and demand forecasts.

### Mobile Application
Develep dedicated mobile apps for NGOs, business owners, and volunteers to make access easier and faster.

### Expand to More Regions
Scale the platform to support more cities and partner with more NGOs and businesses across the country.

### Real-Time Tracking
Integrate live tracking for pickup and delivery to improve transparency and efficiency.

### Automated Impact Reports
Generate detailed reports automatically for NGOs, sponsors and government stakeholders.

### Enhanced Security & Scalability
Improve data security, user authentication, and system scalability to handle a larger user base.

Continuously improving, to feed more people and waste less food.

More Features
Bigger Impact

# Thank You

Thank you for your time and valuable attention.

→ | **Next, we have a live product demonstration.**